

Namal University Mianwali

Department of Electrical Engineering

EE-252L: Introduction to Embedded Systems

Line Following Robot

(A Project of Introduction to Embedded System Semester Course)

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Abstract:

This report presents the design and development of a line following robot using an Arduino Uno microcontroller, DC motors, L298N driver, IR sensors, and wheels. The robot autonomously tracks a black line on a contrasting surface, utilizing an array of four IR sensors for precise movement. The Arduino Uno interface enables algorithm implementation to control motor speed and steering, ensuring smooth movement along the line. This project demonstrates the potential applications of line following robots in industrial automation, household tasks, and tour guidance, highlighting the importance of robotics in various fields

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Introduction:

Line following robots are essential in robotics. They help machines move along specific paths accurately. This project aims to create one using an Arduino Uno microcontroller, DC motors, an L298N driver, IR sensors, and wheels. The robot will follow a visible black line on a different-colored surface by itself. It will use the IR sensors to detect the line and make adjustments to stay on track in real-time. This robot's ability to sense and react to the line could be useful in industries like automation, navigation systems, and automotive tech. The project explores how to make and use a line following robot, showing how effective sensor-based navigation and control systems can be.

Background:

In today's world, technology shapes our lives in significant ways. To make the most of it, we need to understand how to use it and also develop the skills to come up with new ideas. This project is all about learning different things like mechanics, electronics, and programming to create a line following robot. By putting together an Arduino Uno microcontroller, DC motors, an L298N driver, IR sensors, and wheels, we show how hardware and software skills come together. This project helps us think logically, be creative, and solve problems by applying what we've learned in a real-life situation. The robot can follow a path and show how far it's gone on an LCD display, which demonstrates how different skills can work together to make something useful.

Motivation:

We decided to create this line following robot to put into practice what we learned in our embedded systems course. While we understood the theory behind microcontrollers, sensors, and programming, we wanted to go beyond theory and apply our knowledge to a real project. This robot project gave us the chance to do just that. By using our understanding of Arduino Uno, IR sensors, and motor control, we were able to build a working robot. This hands-on experience helped us reinforce what we learned in class, improve our problem-solving abilities, and gain practical skills in designing and building embedded systems. Our goal was to bridge the gap between theory and real-world application, enhancing our understanding of embedded systems and preparing us for future projects and innovations.

Problem Description:

In industrial settings, efficient transportation of products between manufacturing plants and warehouses is crucial. Currently, human-operated carts or trucks are used, leading to reliability and efficiency issues. To address this bottleneck, our project aims to automate this process by designing a line following robot that can transport products between buildings or blocks,

eliminating the need for costly and inconvenient railway tracks. By increasing autonomy and precision, our solution seeks to enhance productivity and reduce labor costs in the industry.

Objectives:

- Design and develop a line following robot using Arduino Uno, DC motors, L298N driver, IR sensors, and wheels.
- The robot must be capable of following a black line on a contrasting surface.
- The robot should be able to take various degrees of turns.
- The robot must be able to adapt to different lighting conditions.

Methodology:

- **Literature Review:** We researched online resources, academic papers, and books to understand embedded systems, line following robots, and Arduino Uno applications.
- **Conceptualization:** We used theoretical concepts of embedded systems to design and develop a line following robot, combining hardware and software components.
- **Component Selection:** We carefully chose appropriate components like Arduino Uno, DC motors, L298N driver, IR sensors, and wheels to ensure they worked well together.
- **Coding and Implementation**: We wrote and debugged code in the Arduino Integrated Development Environment (IDE) to control the robot's movement and sensor interactions.
- **Code Verification**: We tested the code thoroughly using various debugging techniques to make sure the robot could track lines reliably and accurately.
- **Hardware Assembly and Integration:** We put together the hardware components, connecting sensors, motors, and other parts to the Arduino Uno board.
- **Experimental Testing:** We conducted extensive tests to evaluate the robot's performance, accuracy, and adaptability in different environments and lighting conditions.

Limitations:

- Sensitivity to Lighting: The robot's IR sensors might struggle in very bright or dark conditions, which could cause tracking errors or loss of the line.
- Limited Path Adaptability: The robot's algorithm and sensors may not handle complex or irregular paths well, restricting its ability to navigate diverse routes.
- Sensor Calibration Dependency: Precise sensor calibration and alignment are crucial for the robot's performance. Any misalignment or miscalibration could affect its ability to track the line accurately.

Procedure:

In this project, we'll talk about how to make a line-following robot using different parts. First, we'll look at the pieces we need for this project.

Components:

We will discuss the different components of a line-following robot and their roles in helping the robot follow a path accurately.

Arduino Uno Microcontroller:

The Arduino Uno microcontroller acts as the brain of the line-following robot. It processes input from the sensors and sends commands to the motors. This ensures the robot follows the designated path accurately.



Figure 1Arduino Uno Microcontroller

DC Motors:

DC motors drive the wheels of the robot, converting electrical energy into mechanical motion. They are crucial for providing the movement required for navigation. Proper control of these motors is essential for path-following precision.



Figure 2 DC Motor

LN298 Driver:

The L298N driver controls the power delivered to the DC motors. It enables precise regulation of motor speed and direction. This is vital for the accurate movement and steering of the robot.

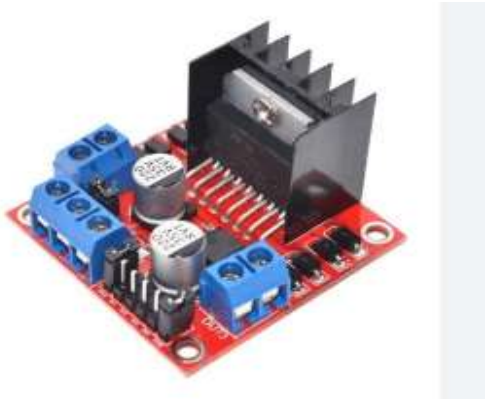


Figure 3 LN298 Driver

IR Sensors:

IR sensors detect the path on the ground, sending critical data to the Arduino Uno. They help the robot determine its position relative to the line. This input is essential for making real-time adjustments to stay on track.



Figure 4 IR Sensors

Wheels:

The wheels and chassis provide the structural support and mobility for the robot. They ensure that all components are securely mounted and enable smooth movement. A well-designed chassis contributes to the robot's stability and performance.



Figure 5 Wheels

Jumper Wires and Switch Button:

Jumper wires create the necessary electrical connections between the various components. The switch button allows the user to easily power the robot on and off. Both elements are essential for the functionality and control of the robot



Figure 6 Jumper wire and Switch button

Circuit Diagram:

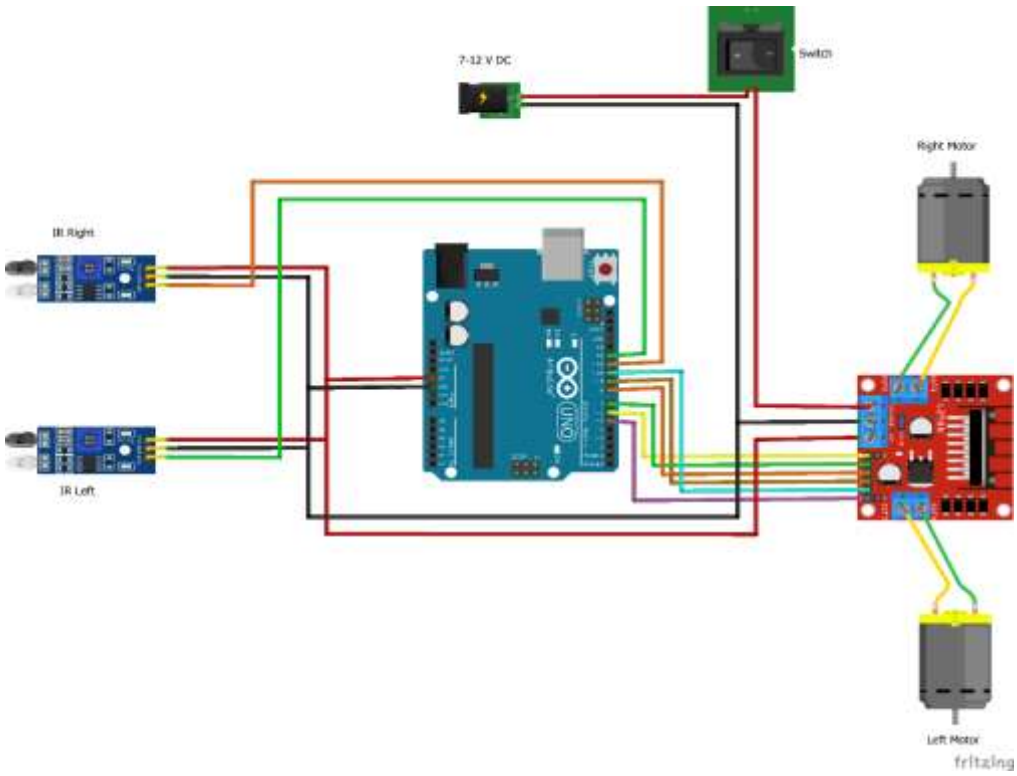


Figure 7 Circuit Diagram of Line Following Robot

Arduino Uno Code:

```
#define IR_SENSOR_RIGHT 11
#define IR_SENSOR_LEFT 12
#define MOTOR_SPEED 180

int enableRightMotor = 6;
int rightMotorPin1 = 7;
int rightMotorPin2 = 8;

int enableLeftMotor = 5;
int leftMotorPin1 = 9;
int leftMotorPin2 = 10;

void setup() {
  TCCR0B = TCCR0B & B11111000 | B00000010;

  pinMode(enableRightMotor, OUTPUT);
  pinMode(rightMotorPin1, OUTPUT);
  pinMode(rightMotorPin2, OUTPUT);

  pinMode(enableLeftMotor, OUTPUT);
  pinMode(leftMotorPin1, OUTPUT);
  pinMode(leftMotorPin2, OUTPUT);
```

```
  pinMode(IR_SENSOR_RIGHT, INPUT);
  pinMode(IR_SENSOR_LEFT, INPUT);
  rotateMotor(0, 0);
}
void loop() {
  int rightIRSensorValue = digitalRead(IR_SENSOR_RIGHT);
  int leftIRSensorValue = digitalRead(IR_SENSOR_LEFT);

  if (rightIRSensorValue == LOW && leftIRSensorValue == LOW) {
    rotateMotor(MOTOR_SPEED, MOTOR_SPEED);
  } else if (rightIRSensorValue == HIGH && leftIRSensorValue == LOW) {
    rotateMotor(-MOTOR_SPEED, MOTOR_SPEED);
  } else if (rightIRSensorValue == LOW && leftIRSensorValue == HIGH) {
    rotateMotor(MOTOR_SPEED, -MOTOR_SPEED);
  } else {
    rotateMotor(0, 0);
  }
}
void rotateMotor(int rightMotorSpeed, int leftMotorSpeed) {
  if (rightMotorSpeed < 0) {
    digitalWrite(rightMotorPin1, LOW);
    digitalWrite(rightMotorPin2, HIGH);
  } else if (rightMotorSpeed > 0) {
    digitalWrite(rightMotorPin1, HIGH);
    digitalWrite(rightMotorPin2, LOW);
  } else {
    digitalWrite(rightMotorPin1, LOW);
    digitalWrite(rightMotorPin2, LOW);
  }

  if (leftMotorSpeed < 0) {
    digitalWrite(leftMotorPin1, LOW);
    digitalWrite(leftMotorPin2, HIGH);
  } else if (leftMotorSpeed > 0) {
    digitalWrite(leftMotorPin1, HIGH);
    digitalWrite(leftMotorPin2, LOW);
  } else {
    digitalWrite(leftMotorPin1, LOW);
    digitalWrite(leftMotorPin2, LOW);
  }
  analogWrite(enableRightMotor, abs(rightMotorSpeed));
  analogWrite(enableLeftMotor, abs(leftMotorSpeed));
}
```

Proteus Circuit Diagram:

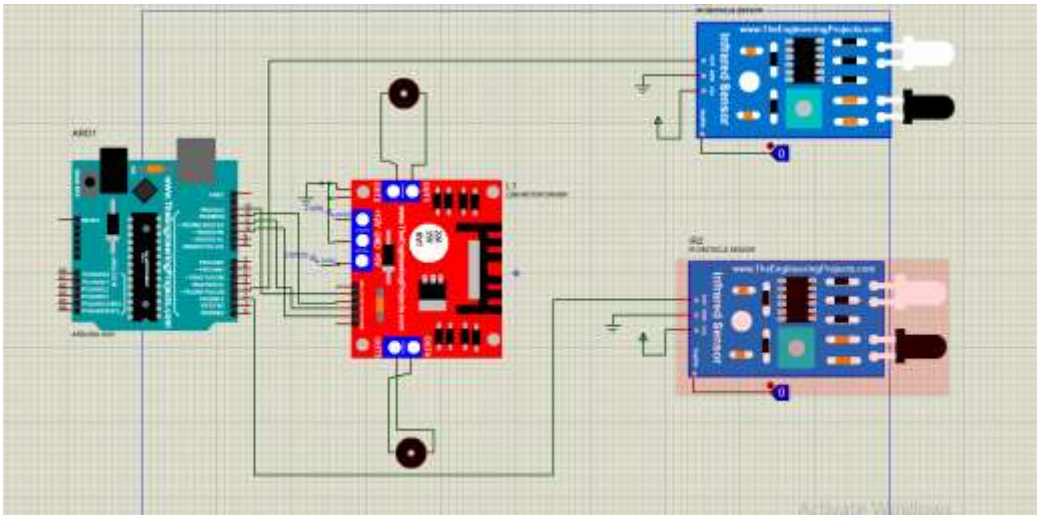


Figure 8 Proteus Circuit Diagram

Path which robot followed:

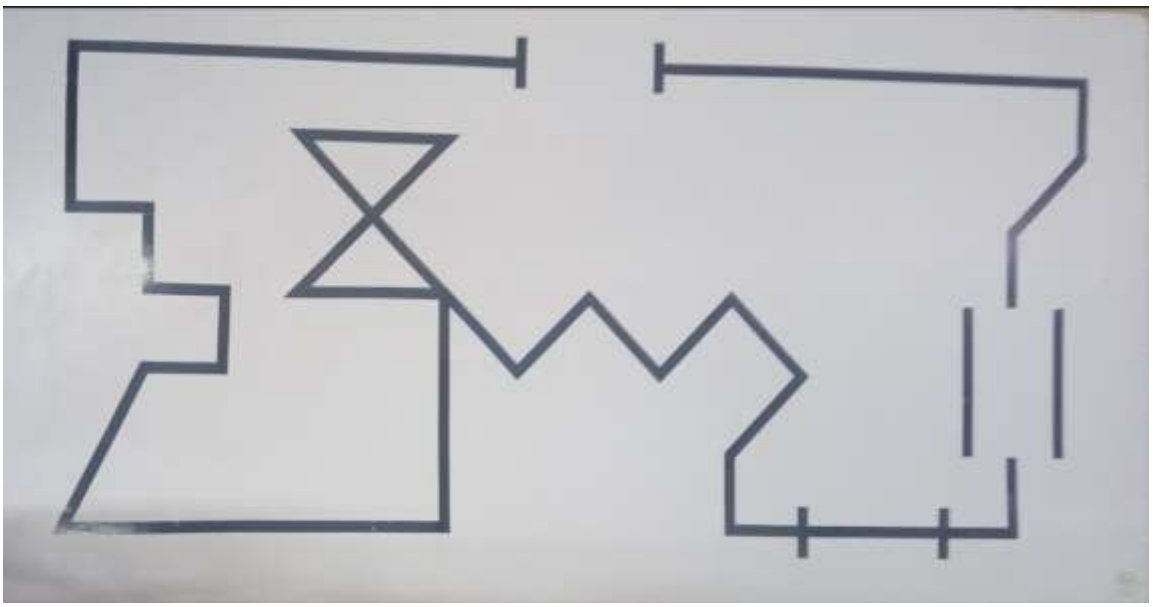


Figure 9 Path which robot followed:

Conclusion:

we built a robot that can follow lines using basic components like Arduino Uno, L298 motor driver, IR sensors, motors, wheels, and switches. It can detect and follow a black line on a white surface, adjusting its path as needed.

We learned a lot about how things like electronics, mechanics, and programming can work together to create something cool. We also realized how important teamwork and communication are when trying out new ideas.

Our little line-following robot demonstrates how combining different skills can lead to practical solutions.

References:

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